Physics - Undergraduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
   (a) Magnetic susceptibility
   (b) Hall coefficient
   (c) Electrical resistivity
   (d) Thermal conductivity

2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
   (a) 2
   (b) 250
   (c) 5,000
   (d) 10,000

3. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
   (a) As
   (b) P
   (c) Sb
   (d) B

4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T. The average total energy of the oscillator is
   (a) (1/2) k T
   (b) kT
   (c) (3/2) k T
   (d) 3 kT

5. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
   (a) 44 Hz
   (b) 196 Hz
   (c) 262 Hz
   (d) 393 Hz

True / False (10 marks)

Answer True or False

6. True or False: A particle moving at v=0.60 c in the lab frame will travel 450 m before decaying.
7. True or False: The eigenvalues of a Hermitian operator can be imaginary in certain cases.
8. True or False: The resolving power of the spectrometer for the given wavelengths is 250.
9. True or False: The energy required to ionize helium by removing one electron is 39.5 eV.
10. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately 1.0 GeV/c^2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A student measures the disintegration of a radioactive isotope over 10 one-second intervals, obtaining values of 3, 0, 2, 1, 2, 4, 0, 1, 2, and 5. Analyze how long the student should count to achieve a measurement uncertainty of 1 percent, and explain the factors influencing this determination.
12. Discuss the advantages of using a dye laser for spectroscopy over a range of visible wavelengths, and analyze its effectiveness compared to other laser types.

End of Paper